

Hybrid surrogate modeling with Physically Recurrent Neural Networks

Iuri Rocha

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High-performance and complex materials

We rely on high-performance materials for many applications, and that reliance will only increase

- Maximum efficiency with minimum material use, high tailorability
- Urgent challenges in climate adaptation, durable infrastructure, sustainable structures



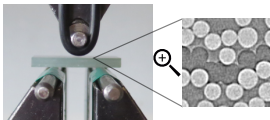
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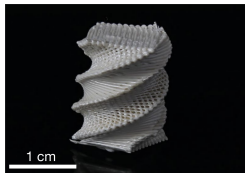
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At the same time, whole new families of materials are emerging

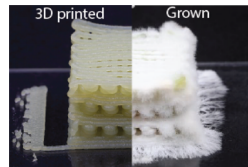
- 3D printing with molecular-scale control, engineered living materials



[Rocha *et al.* (2019), **EJMA**]



[Gantenbein *et al.* (2018), **Nature**]



[Gantenbein *et al.* (2023), **Nature Materials**]

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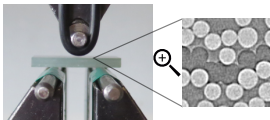
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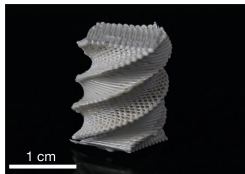
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How can we better understand current materials and design new ones?

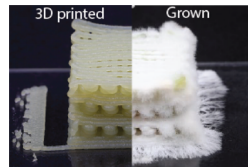
- A purely experimental approach is in any case a no go



[Rocha et al. (2019), **EJMA**]



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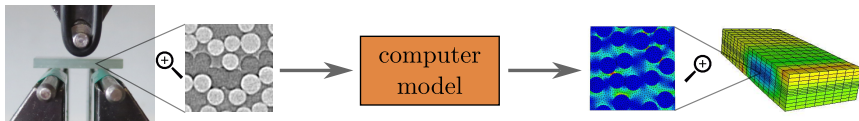
[Gantenbein et al. (2023), **Nature Materials**]

Materials and computer models – avenues for progress

The answer: combining a handful of experiments with extensive **virtual testing**

Multiscale modeling

- Model materials at very small and very large scales at the same time
- More knowledge $\Rightarrow$ smaller safety factors $\Rightarrow$ **more efficient designs**



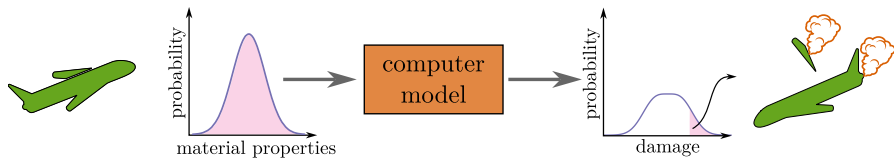
The challenge: multiscale modeling requires an immense amount of microscale computations

Materials and computer models – avenues for progress

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Uncertainty Quantification

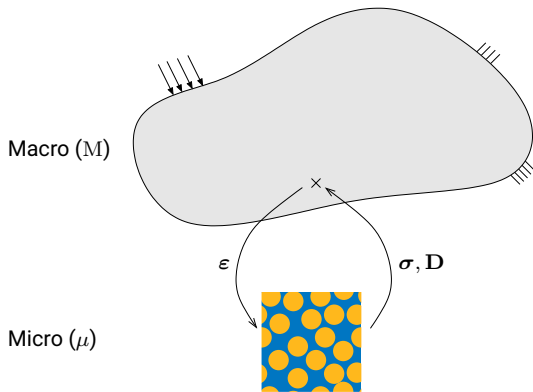
- Material properties and some service loads are inherently uncertain
- Better understanding of their impact on performance $\Rightarrow$ **more efficient designs**



The challenge: UQ often requires an immense amount of model evaluations

For now let us focus on multiscale simulations

Compute stress σ as the volume average in a deformed Representative Volume Element (RVE)



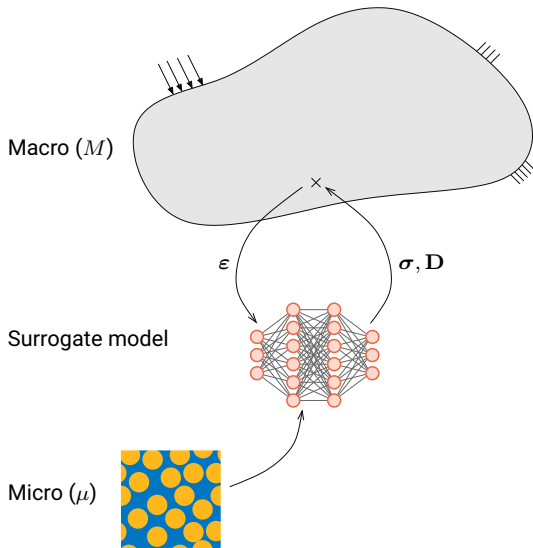
The FE² method:

- True separation of scales is supported
- No macroscale constitutive assumptions
- Compatible with regular FEM workflow

Challenge:

- **Extreme** computational cost

The constitutive model can be replaced with a data-driven surrogate

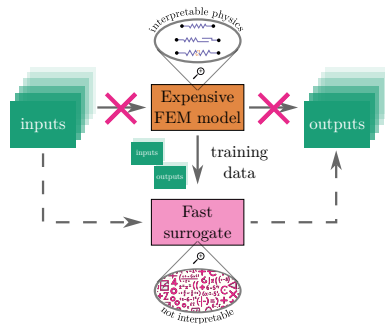


The usual approach:

- Run a few full-order simulations
- Use the fast surrogate online

Challenge:

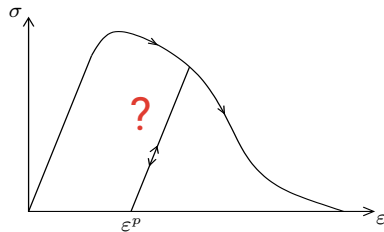
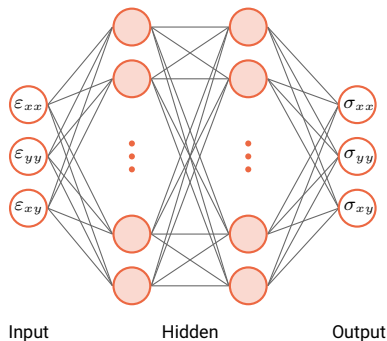
- Physics is completely discarded



Neural network surrogates for multiscale

Neural nets are universal approximators, so they are a popular choice:

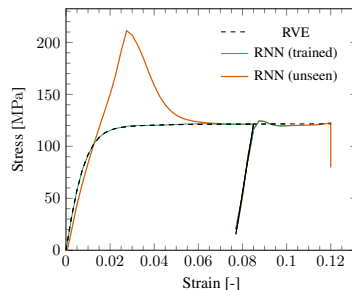
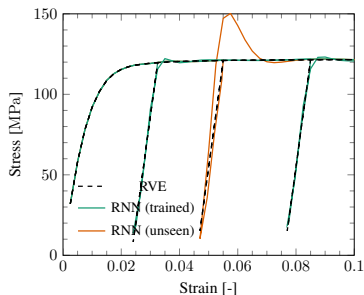
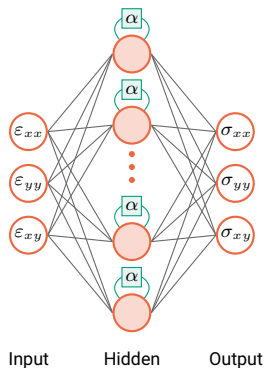
- Conventional networks can map strains to stresses but cannot handle path dependence



Neural network surrogates for multiscale

Neural nets are universal approximators, so they are a popular choice:

- Conventional networks can map strains to stresses but cannot handle path dependence
- Recurrent neural networks solve that issue but can be a nightmare to train

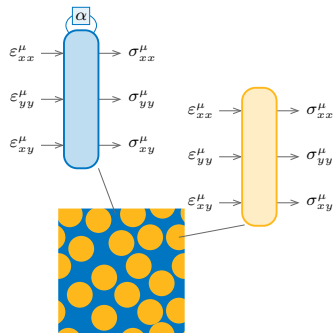
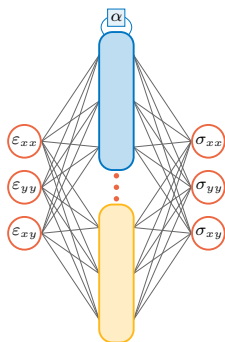


[Rocha, Kerfriden, van der Meer (2023), **Mech Mater**]

Physically Recurrent Neural Networks

A hybrid machine learning approach:

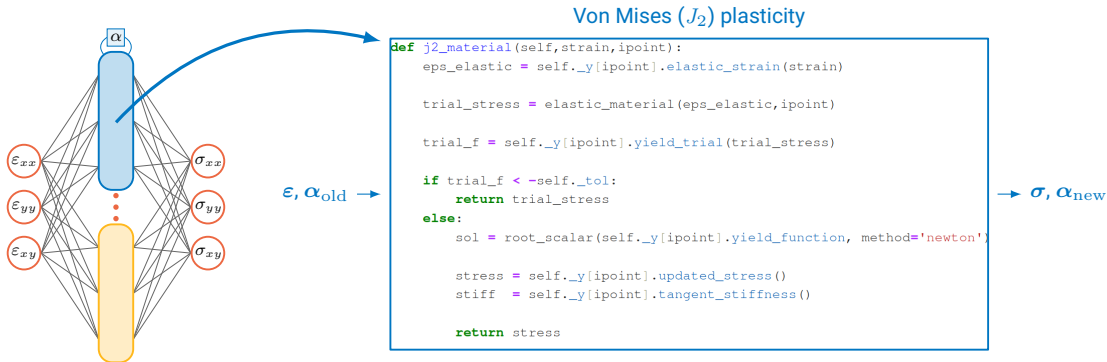
- Embed actual material models inside the network, no need to learn time dependence again
- Which models to add? Easy choice: the ones in the micromodel we are substituting



Physically Recurrent Neural Networks

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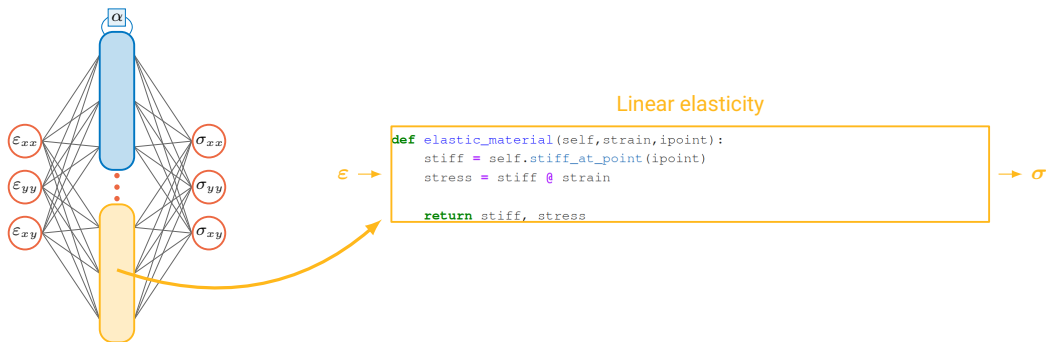
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Physically Recurrent Neural Networks

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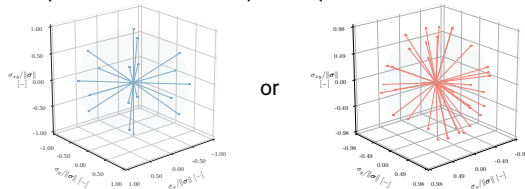


PRNNs versus conventional ML models

Training and testing with the same type of data:

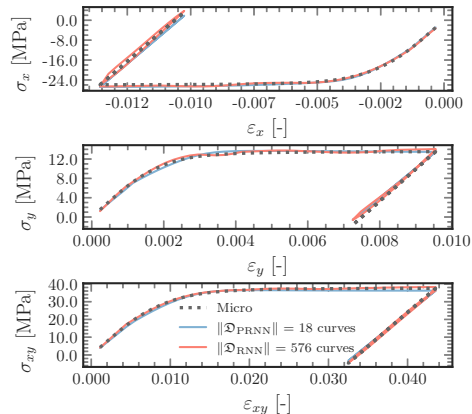
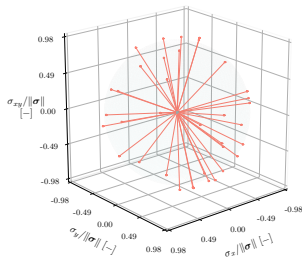
PRNN (18 monotonic curves) RNN (576 curves with unloading)

Train



or

Test

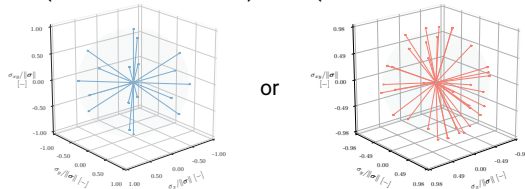


PRNNs versus conventional ML models

Testing with more challenging paths:

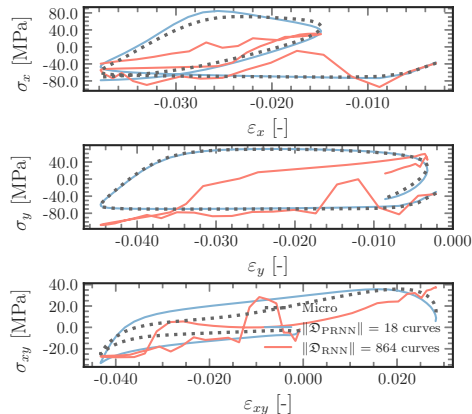
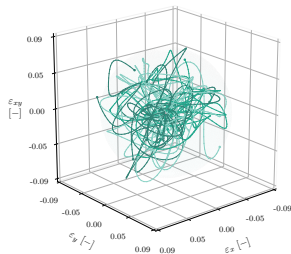
PRNN (18 monotonic curves) RNN (864 curves with unloading)

Train



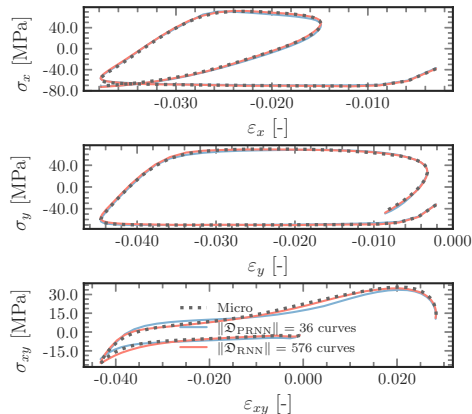
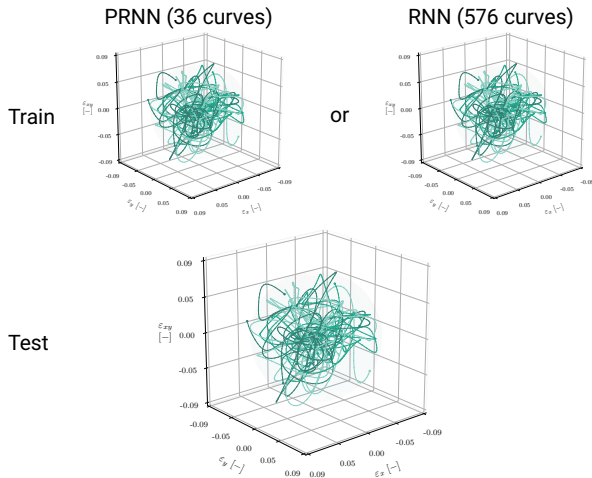
or

Test



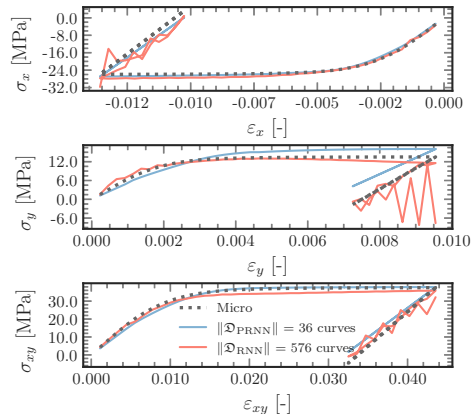
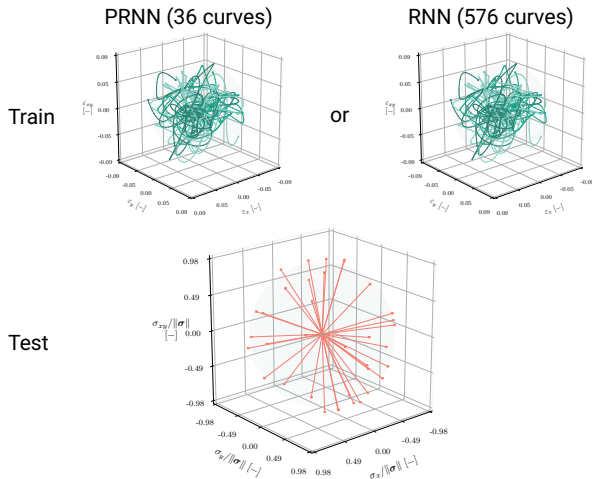
PRNNs versus conventional ML models

Okay, how about training with the challenging paths?



PRNNs versus conventional ML models

Great, but now we can do the simple ones again... right? **No!**

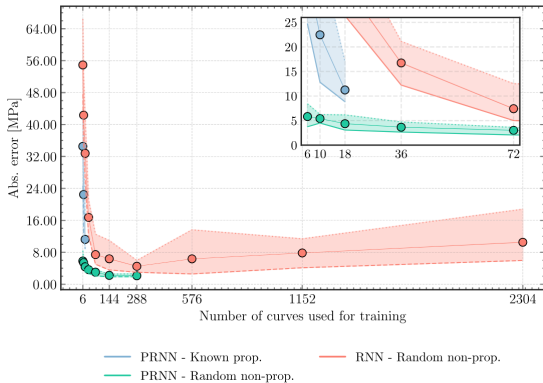


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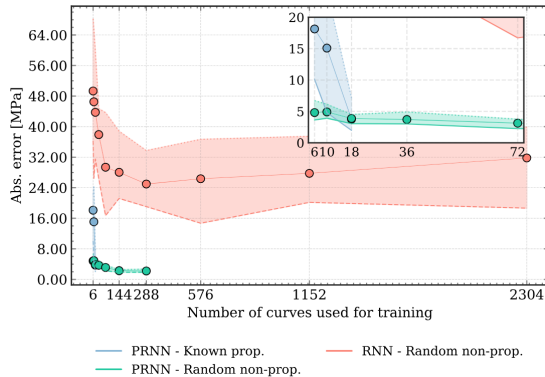
Learning curves with 10 realizations per dataset size:

- PRNNs remain accurate even when trained with very small datasets

Test set with 100 GP paths



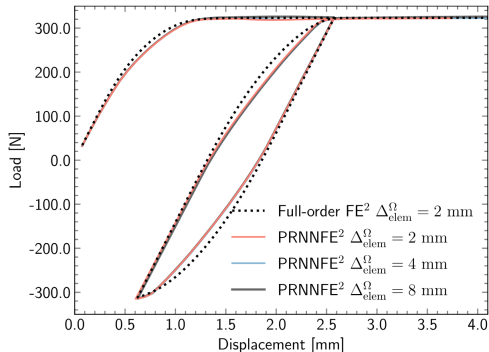
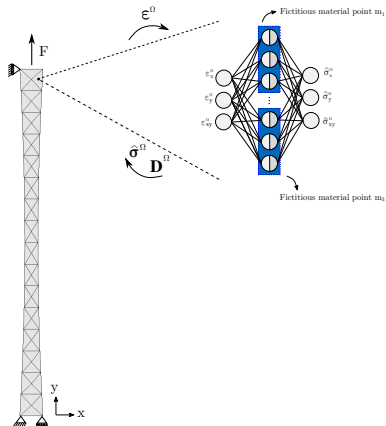
Test set with 100 monotonic paths ($10\times$ smaller Δt)



PRNNs for FE^2

Trying out the model for FE^2 , tapered bar with different mesh densities:

- Speed-ups against FE^2 of $20000\times \sim 30000\times$
- Trained only on monotonic paths!

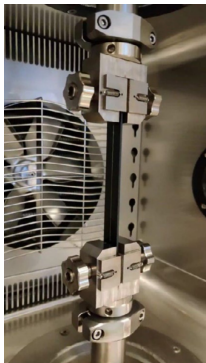


What we are now able to do: FE² versus experiments

We also used it to confirm some experimental results that had been puzzling us:

- Uniaxial tension on off-axis carbon-PEEK coupons

Uniaxial tension experiment



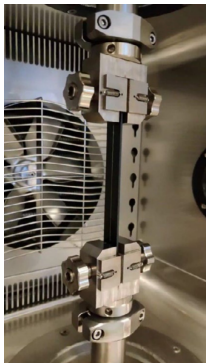
[Maia, Rocha, van der Meer (2025), [arXiv:2501:10193](#)]

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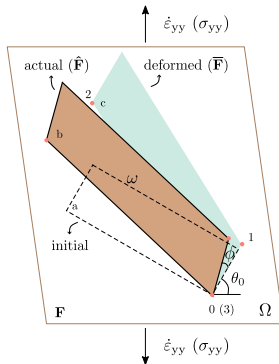
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Specialized arc-length



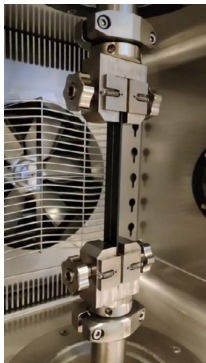
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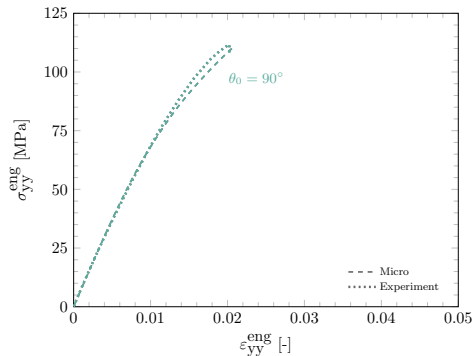
Uniaxial tension experiment



Off-axis angle



Engineering stress-strain

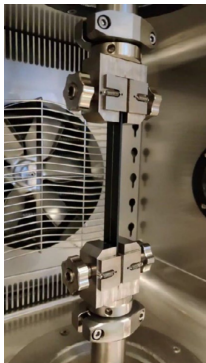


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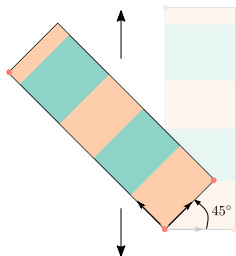
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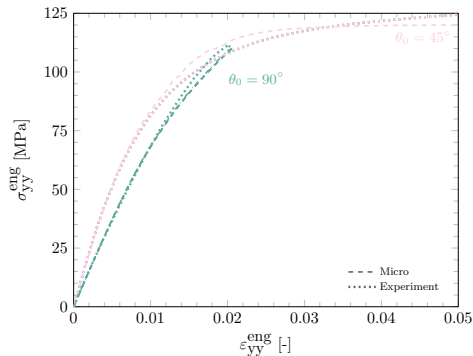
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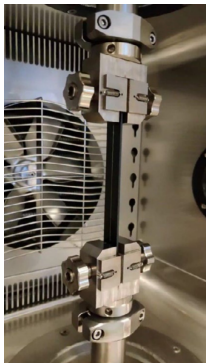
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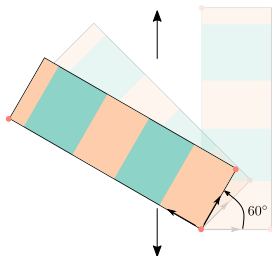
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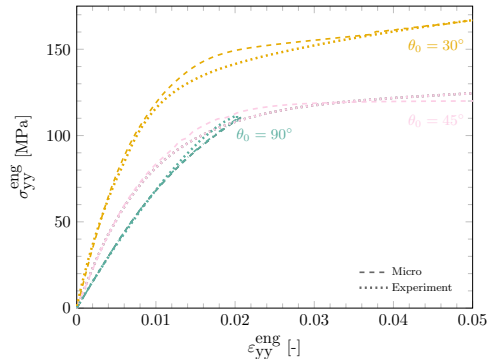
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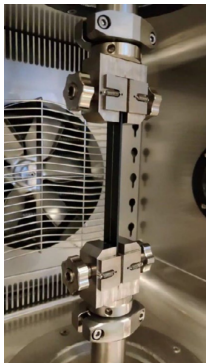


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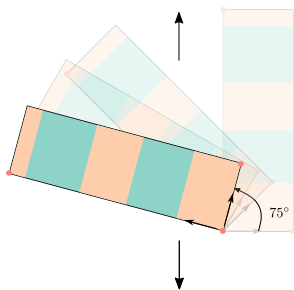
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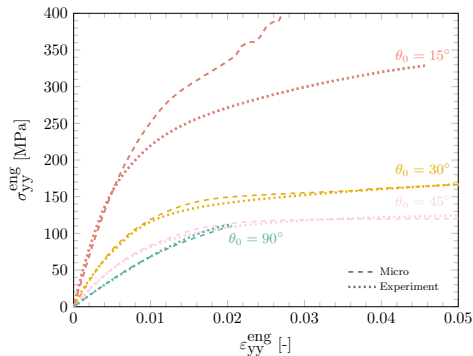
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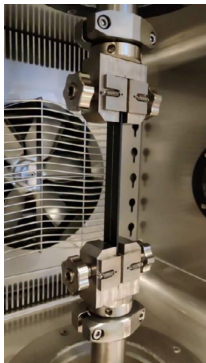


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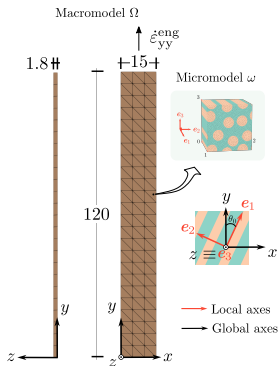
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Uniaxial tension experiment



Multiscale model



Engineering stress-strain

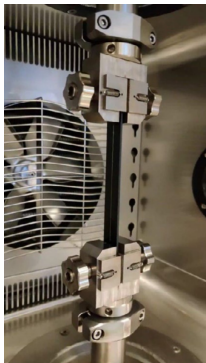
too expensive
to simulate

What we are now able to do: FE² versus experiments

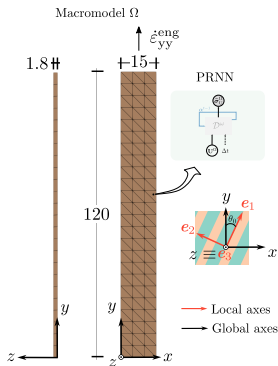
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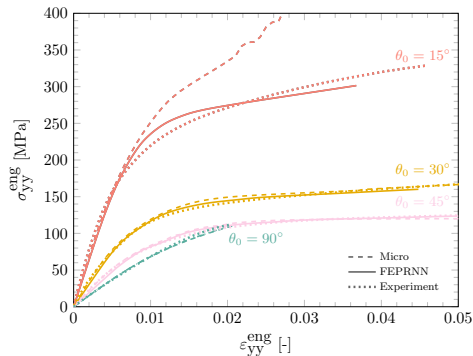
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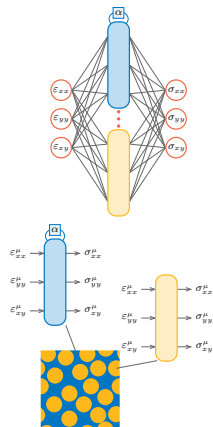


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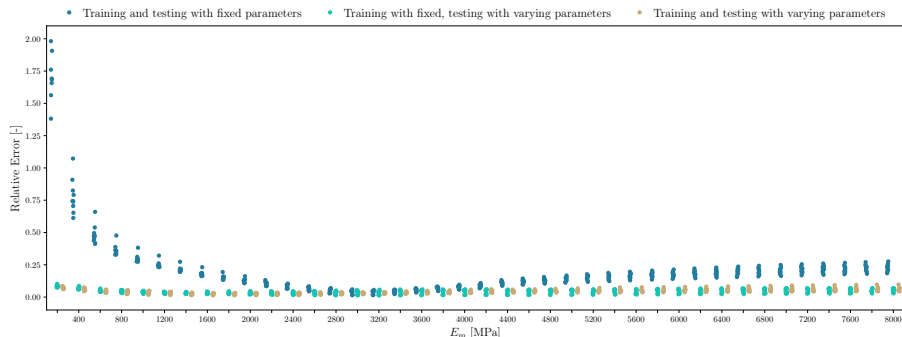
What we are now able to do: Microscale UQ

The network inherits material properties. Can we also extrapolate for those?

- Apparently yes! Train with a single value, extrapolate quite far



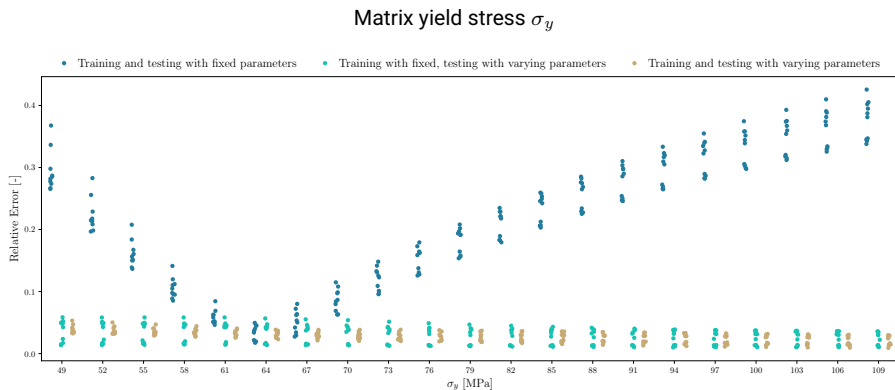
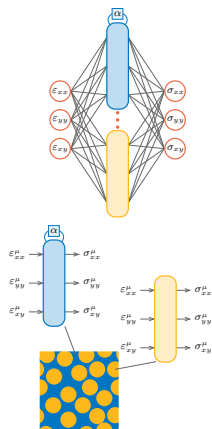
Matrix Young's modulus E_m



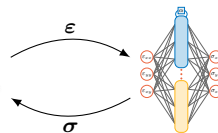
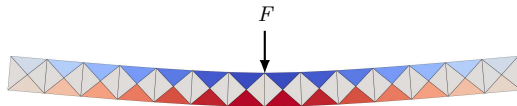
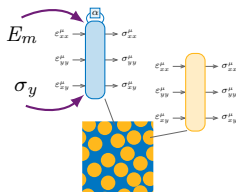
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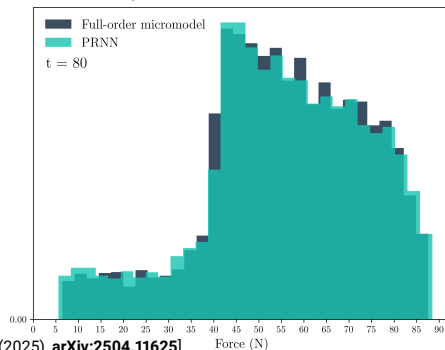
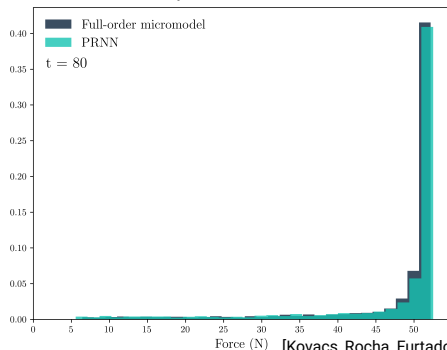


What we are now able to do: Multiscale UQ



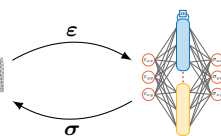
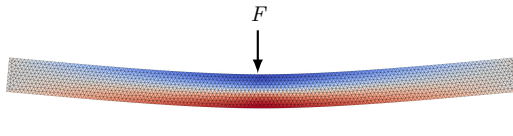
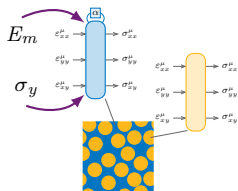
$$p(F) = \int p(F|E_m)p(E_m) dE_m$$

$$p(F) = \int p(F|E_m, \sigma_y)p(E_m, \sigma_y) dE_m d\sigma_y$$



[Kovacs, Rocha, Furtado, Camanho, van der Meer (2025), [arXiv:2504.11625](https://arxiv.org/abs/2504.11625)]

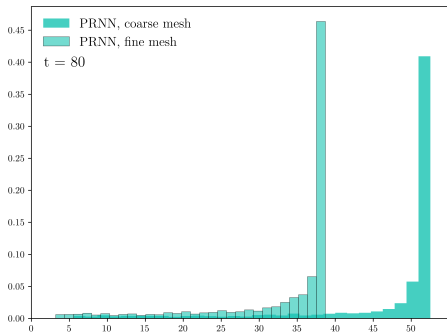
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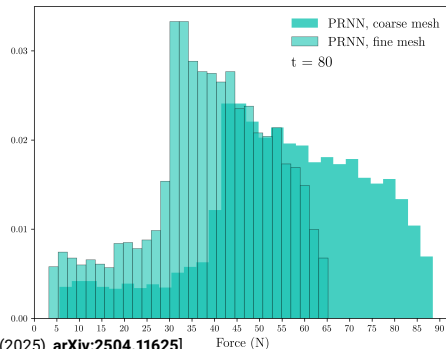
Three-point bending, *fine* macro mesh:

- Monte Carlo forward UQ, $N = 5000$ samples
- PRNN trained with a single (E_m, σ_y) pair
- Sequential FE² UQ would now take ≈ 50 years to run

$$p(F) = \int p(F|E_m)p(E_m) dE_m$$



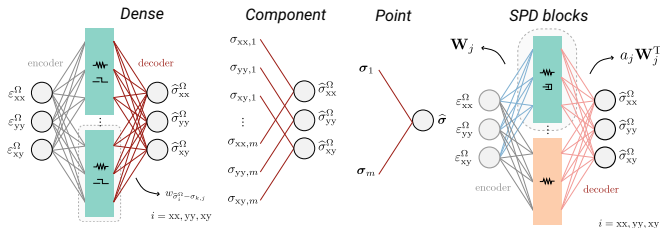
$$p(F) = \int p(F|E_m, \sigma_y)p(E_m, \sigma_y) dE_m d\sigma_y$$



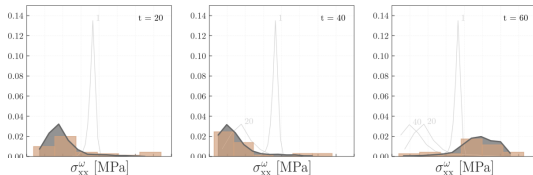
But wait, what about thermodynamics!?

Architecture flexibility allows for a lot of experimentation:

- Sparse decoder connectivities improve interpretability and local stress distributions
- SPD encoder/decoders with shared weights guarantee thermodynamic consistency

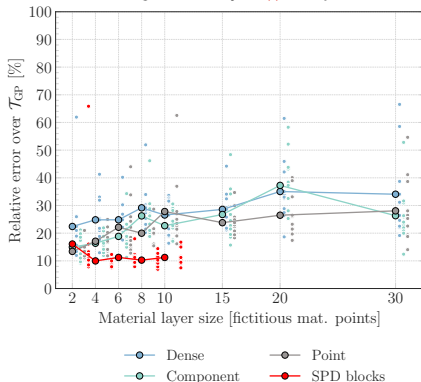


Stress histograms for a random GP path (Component architecture)



[Maia, Rocha, Meyer, van der Meer (2025), In preparation]

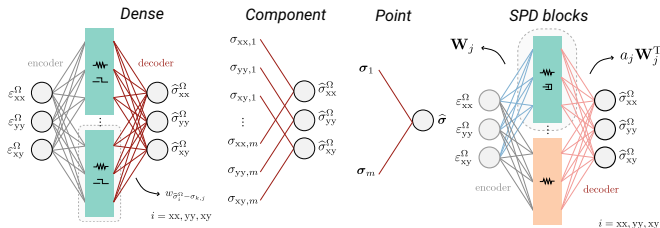
Training with only 2 (!) GP paths



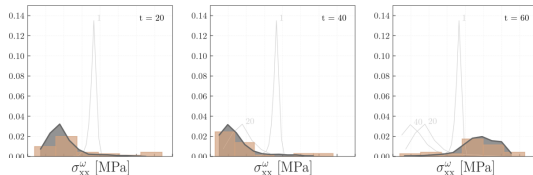
But wait, what about thermodynamics!?

Architecture flexibility allows for a lot of experimentation:

- Sparse decoder connectivities improve interpretability and local stress distributions
- SPD encoder/decoders with shared weights guarantee thermodynamic consistency

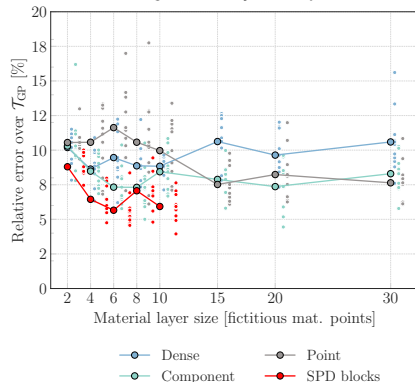


Stress histograms for a random GP path (Component architecture)



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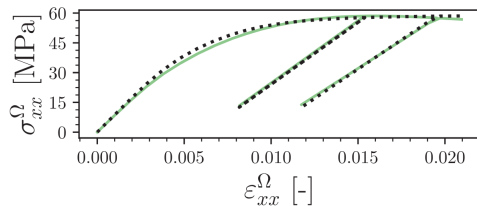
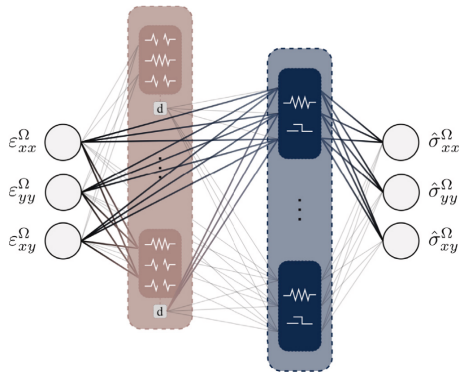
Training with only 5 GP paths



But wait, what about failure!?

Matrix plasticity combined with fiber/matrix debonding

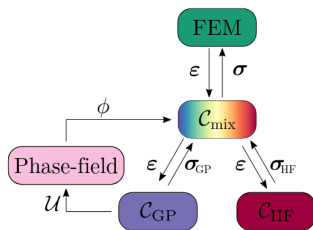
- Two types of embedded material models, extra architecture tricks
- Combination of stiffness loss and plastic strains well captured



Other recent works in the lab: Phase fields for surrogate modeling

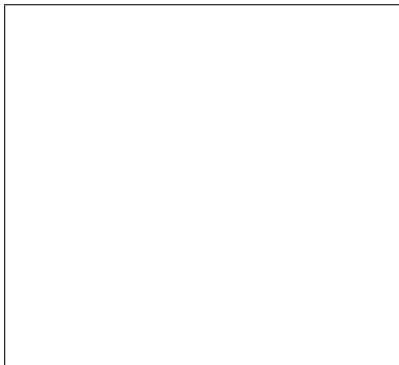
Using a phase field potential to drive active learning:

- Smooth mix of high-fidelity and surrogate models for numerical stability



→ only high-fidelity

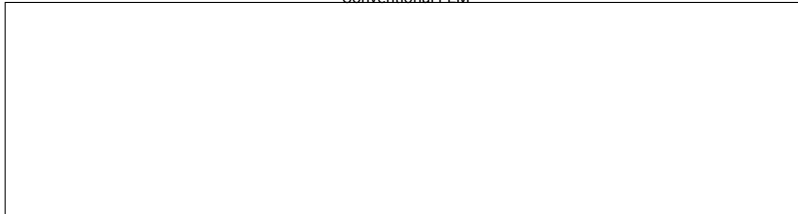
→ only surrogate



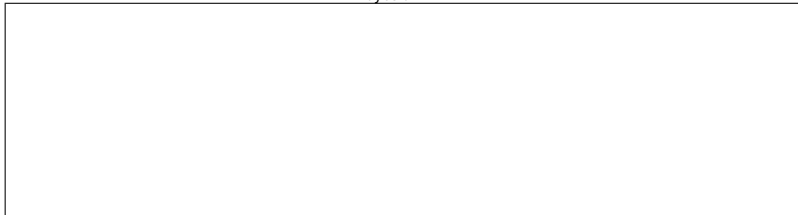
Other recent works in the lab: Bayesian FEM

Assume a prior over the solution on a fine mesh, condition on shape functions from a coarse mesh:

Conventional FEM



Bayesian FEM

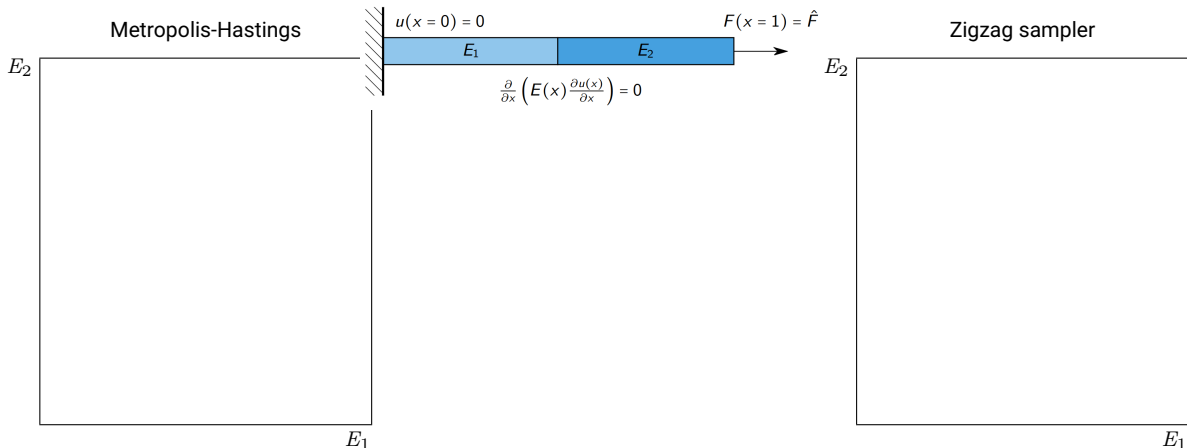


[Poot, Kerfriden, Rocha, van der Meer (2025), [arXiv:2506.02815](#)]

Other recent works in the lab: Zigzag samplers

Efficient Bayesian inverse modeling through Piecewise Deterministic Markov Processes:

- High-efficiency gradient-informed sampling



[Riccius, Rocha, Bierkens, Kekkonen, van der Meer, **In preparation**]

Thank you!

Feel free to check our website:

- Links to all of our code repositories
- Links to papers and preprints
- Slides from talks, including this one ;)

We are looking for interesting applications!

